MOS Capacitor

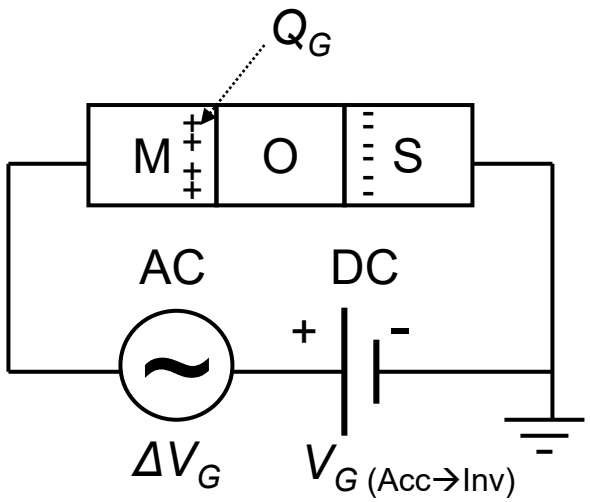
- Introduction
- Ideal MOS Capacitors
  - Qualitative Description of MOS Capacitor
  - Quantitative Analysis of MOS Capacitor
- *C-V* Characteristics of MOS Capacitor
- Non Ideal MOS-Capacitor
  - Defect Charges in MOS Capacitors
- Applications of *C-V* curves

C-V Characteristics of MOS Capacitor

- Small signal capacitance

$$C \equiv \frac{\Delta Q_G}{\Delta V_G}$$

$$\frac{1}{C} = \frac{\Delta V_G}{\Delta Q_G} = \frac{\Delta V_{OX}}{\Delta Q_G} + \frac{\Delta \Phi_S}{\Delta Q_G} = \frac{1}{C_{OX}} + \frac{1}{C_{Si}}$$



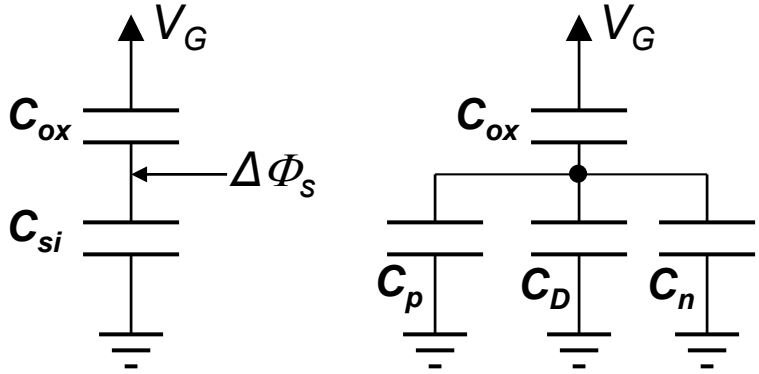
Substrate capacitance C_{Si}

$$C_{Si} = -\frac{\Delta Q_S}{\Delta \Phi_S} = -\frac{\Delta Q_P}{\Delta \Phi_S} - \frac{\Delta Q_D}{\Delta \Phi_S} - \frac{\Delta Q_n}{\Delta \Phi_S}$$

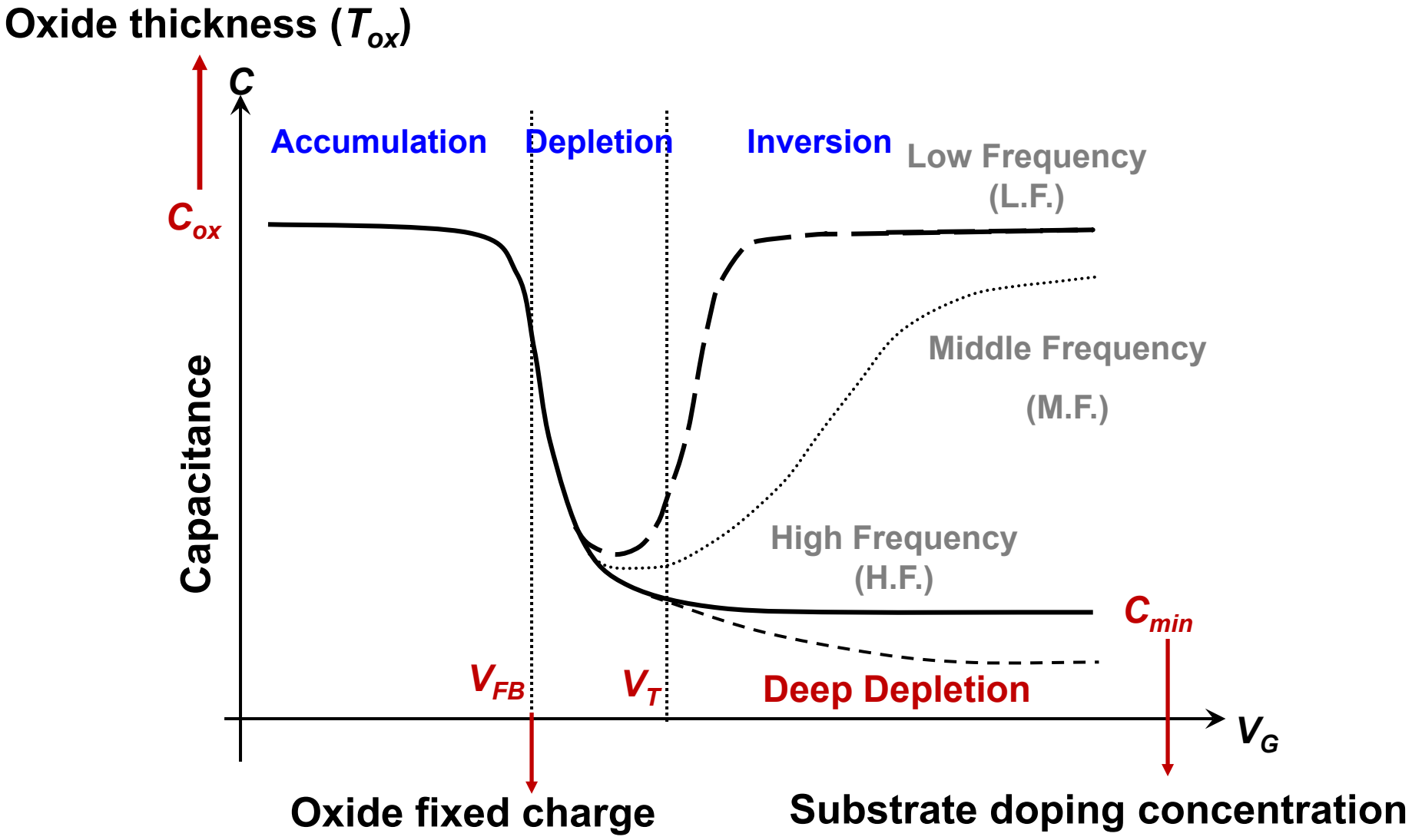
$$C_{Si} = C_P + C_D + C_n$$

DC : DC bias scanning rate

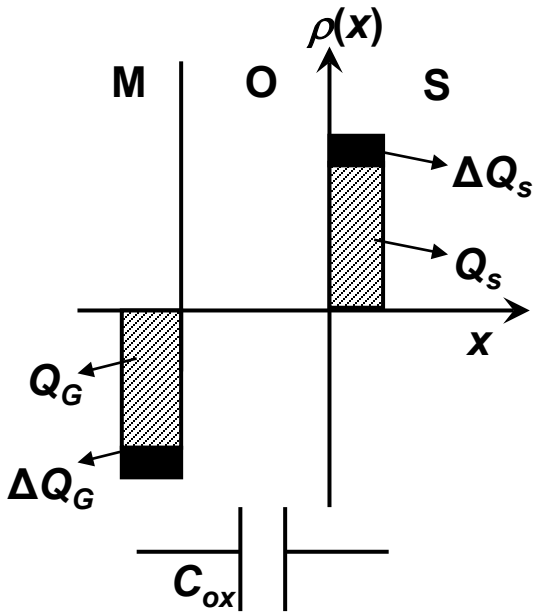
AC : Frequency of small signal



C-V Curve


Oxide thickness (T_{ox})
C
C_{ox}
Capacitance
Accumulation
Depletion
Inversion
Low Frequency
(L.F.)
Middle Frequency
(M.F.)
High Frequency
(H.F.)
C_{min}
V_{FB}
V_T
Deep Depletion
V_G
Oxide fixed charge
Substrate doping concentration

Accumulation



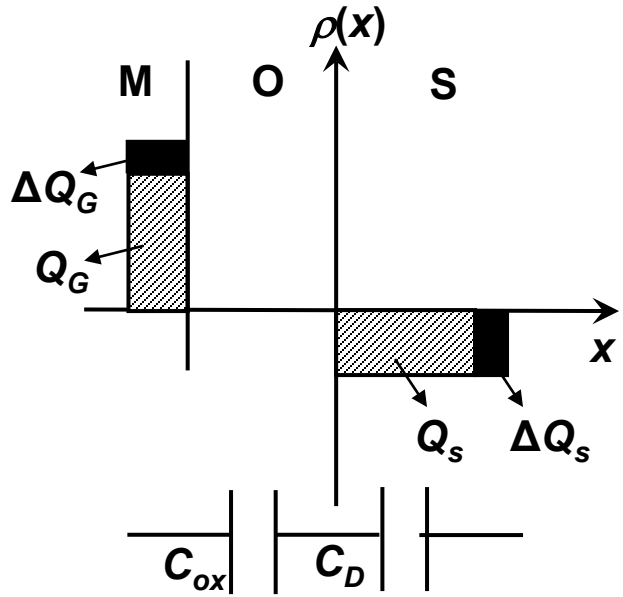
Accumulation

- Charge responding to the AC signal is the accumulation charge (majority carrier)

  → $C = C_{ox}$, independent of AC signal frequency.

  where $C_{ox} \propto 1/T_{ox}$

Depletion



Depletion

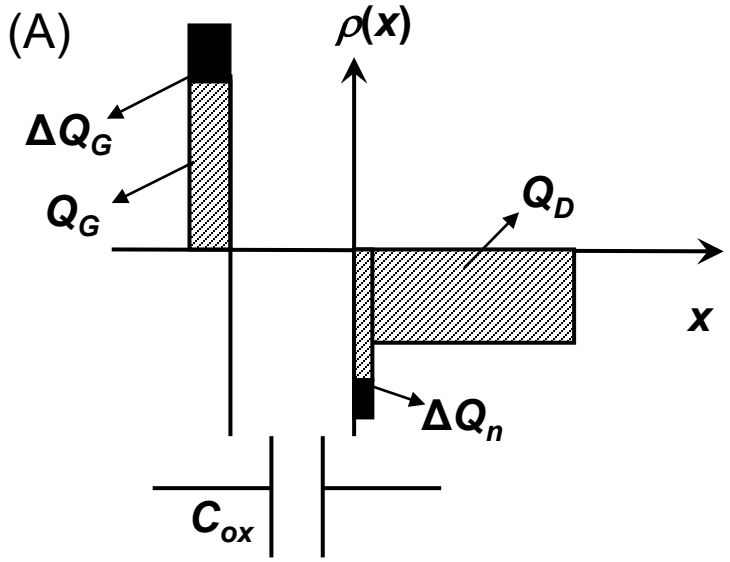
- Charge responding to the AC signal is the depletion charge (majority carrier)

  → $C = C_{ox}C_D / (C_D+C_{ox})$, which is independent of AC signal frequency.

  Where $C_{ox} \propto 1/T_{ox}$, $C_D \propto 1/W$

Inversion

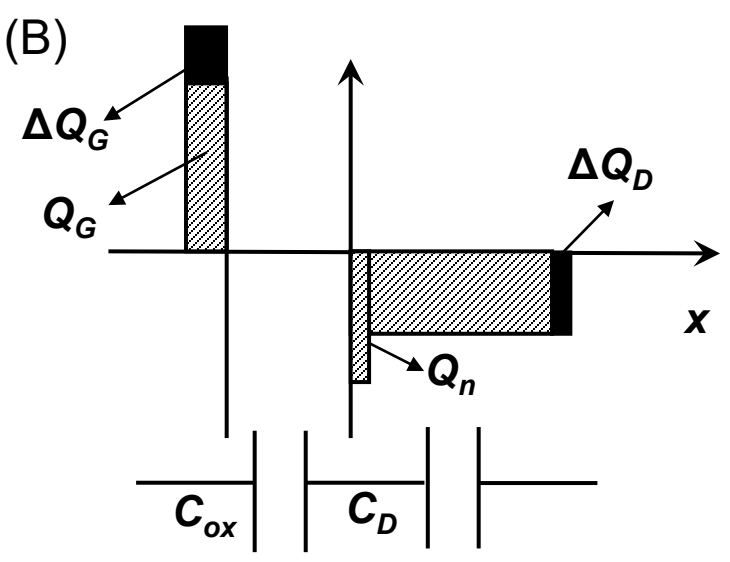
- Capacitance and charge responding to the ac signal are dependent upon the frequency, light, and temperature.
- Characteristics are related to the minority carrier response time.



(A) Low-Frequency

- Frequency is low enough for inversion charge to respond.
- Responding charge is inversion charge

→ $C = C_{ox}$

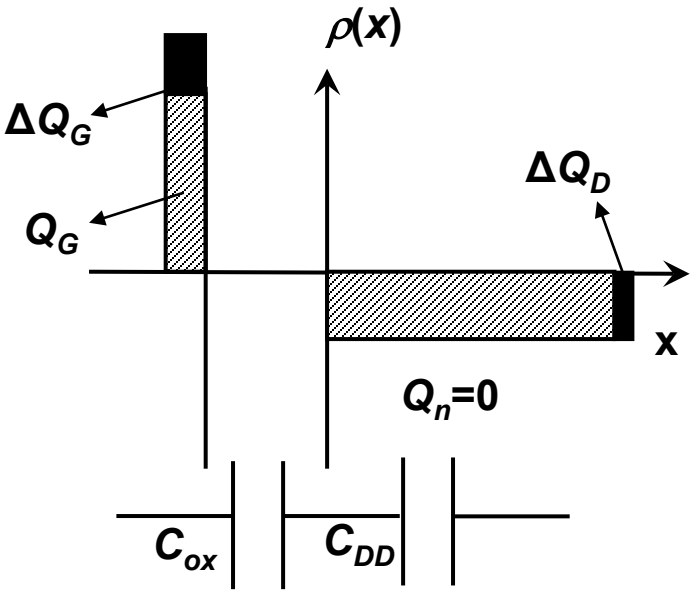


(B) High-Frequency

- Frequency is too high for inversion charge to follow
- Responding charge is depletion

→ $C = C_{min}$

Deep Depletion



- Scan rate is too fast for inversion change to form: $Q_n = 0$
- Additional V_G beyond V_T contributes to Q_D and depletion width extends beyond W_m:
  → $W_D > W_m$, $C_D = \varepsilon_{si} / W_m$
  $C = C_{ox}C_D / (C_D + C_{ox})$,
- ξ_{max} increase until a point when ξ_{max} reaches ξ_{crit}, where Si breakdown

$$V_{G,BD} = V_{FB} + \Phi_s + \frac{Q_{D,BD}}{C_{ox}}$$

$$Q_{D,BD} = q \cdot N_A \cdot W_D = \varepsilon_{Si} \cdot \xi_{crit}$$

C-V from Depletion Approximation

Depletion

Substrate depletion layer capacitance

$$C_{Si} = -\frac{\partial Q_S}{\partial \Phi_S}$$

by depletion approximation

$$Q_S = -qN_A W = -\sqrt{2qN_A \varepsilon_{si} \Phi_S}$$

$$C_{Si} = \sqrt{\frac{qN_A \varepsilon_{si}}{2\Phi_s}}$$

Total capacitance

$$C_T = \frac{C_{ox} \cdot C_{si}}{C_{ox} + C_{si}}$$

$$\frac{C_T}{C_{ox}} = \frac{1}{1 + \frac{C_{ox}}{C_{si}}} = \frac{1}{1 + C_{ox}\sqrt{\frac{2\Phi_s}{qN_A \varepsilon_{si}}}}$$

$$V_G = V_{ox} + \Phi_s = -\frac{Q_s}{C_{ox}} + \Phi_s$$

(1) $$V_G = \frac{\sqrt{2qN_A \varepsilon_{si}}}{C_{ox}} \cdot \sqrt{\Phi_s} + \Phi_s$$

$$\left(X = \sqrt{\Phi_s} \qquad \alpha = \frac{\sqrt{2qN_A \varepsilon_{si}}}{C_{ox}} \right)$$

Equation (1) becomes $$X^2 + \alpha X - V_G = 0$$

$$\sqrt{\Phi_s} = X = \frac{-\alpha + \sqrt{\alpha^2 + 4V_G}}{2}$$

$$\sqrt{\Phi_s} = -\frac{1}{C_{ox}}\sqrt{\frac{qN_A \varepsilon_{si}}{2}} + \sqrt{\frac{2qN_A \varepsilon_{si}}{C_{ox}^2} + 4V_G}$$

$$\frac{C_T}{C_{ox}} = \frac{1}{\sqrt{1 + \frac{2\varepsilon_{ox}^{\ 2}}{qN_A \varepsilon_{si} t_{ox}^2} \cdot V_G}}$$

Inversion

On the onset of inversion

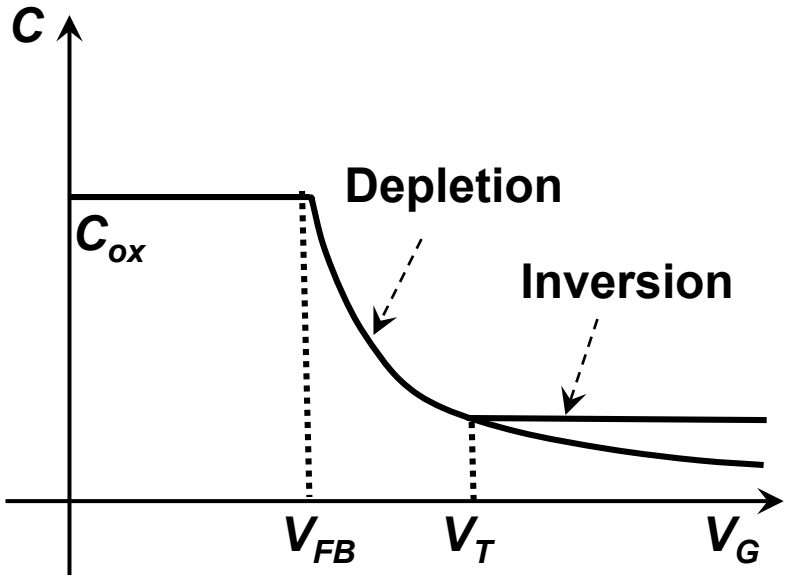
$$\Phi_s = 2\Phi_B \quad \textbf{where} \quad \Phi_B = \frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right)$$

$$W_m = \sqrt{\frac{2\varepsilon_{Si}\Phi_s}{qN_A}} \approx \sqrt{\frac{2\varepsilon_{Si}}{qN_A}\cdot(2\Phi_B)}$$

$$C_{si} = -\frac{\partial Q_s}{\partial \Phi_s} \qquad Q_s = -qN_A W \qquad C_T = \frac{C_{ox}\cdot C_{si}}{C_{ox}+C_{si}}$$

$$\frac{C_T}{C_{ox}} = \frac{1}{1+\frac{C_{ox}}{C_{si}}} = \frac{1}{1+C_{ox}\frac{W_m}{\varepsilon_{si}}}$$

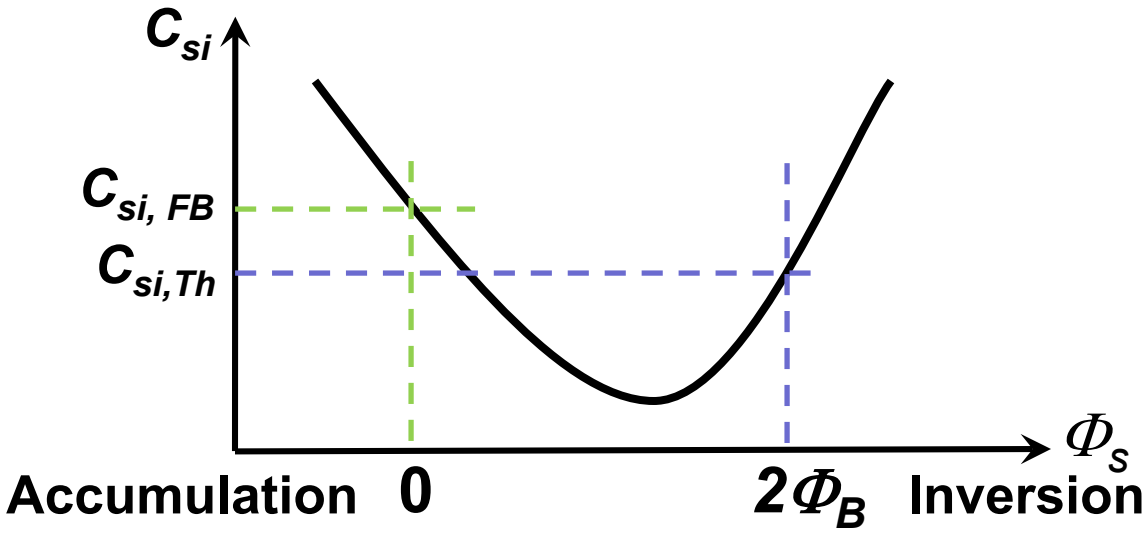
$$= \frac{1}{1+C_{ox}\cdot\sqrt{\frac{4\Phi_B}{qN_A\varepsilon_{si}}}}$$



C-V from General Analysis

$$Q_s = \pm\sqrt{\frac{2q\varepsilon_{si}}{\beta}}\cdot\left[p_{p0}(e^{-\beta\Phi_s}+\beta\Phi_s-1)+n_{p0}(e^{\beta\Phi_s}-\beta\Phi_s-1)\right]^{\frac{1}{2}}$$

$$C_{si}\equiv\frac{\partial Q_s}{\partial\Phi_s}=\sqrt{\frac{2q\varepsilon_{si}}{\beta}}\cdot\frac{\left[p_{p0}(-\beta e^{-\beta\Phi_s}+\beta)+n_{p0}(\beta e^{\beta\Phi_s}-\beta)\right]}{2\cdot\left[p_{p0}(e^{-\beta\Phi_s}+\beta\Phi_s-1)+n_{p0}(e^{\beta\Phi_s}-\beta\Phi_s-1)\right]^{\frac{1}{2}}}$$



At Flat band

$$\Phi_s = 0, C_{si} = \frac{0}{0}$$

For $\Phi_s \rightarrow 0$, and

using Taylor Series Expansion

$$e^{\beta\Phi} = 1 + \beta\Phi + \frac{(\beta\Phi)^2}{2} + \ldots$$

$$e^{-\beta\Phi} = 1 - \beta\Phi + \frac{(\beta\Phi)^2}{2} + \ldots$$

Inserting these into

$$\frac{d\Phi}{dx} = -\sqrt{\frac{2q}{\beta\varepsilon_{si}}}[p_{p0}(e^{-\beta\Phi} + \beta\Phi - 1) + n_{p0}(e^{\beta\Phi} - \beta\Phi - 1)]^{1/2}$$

$$\frac{d\Phi}{dx} \approx -\sqrt{\frac{2q}{\beta\varepsilon_{si}}}[p_{p0} \cdot \frac{(\beta\Phi)^2}{2} + n_{p0} \cdot \frac{(\beta\Phi)^2}{2}]^{1/2}$$

$$(\ p_{p0} \approx N_A >> n_{p0})$$

Therefore, $$\frac{d\Phi}{dx} \approx -\sqrt{\frac{qN_A\beta}{\varepsilon_{si}}}\Phi$$

Boundary Condition:

$$\Phi(x=0) = \Phi_s, \Phi(x=\infty) = 0$$

$$\Rightarrow \Phi(x) = \Phi_s \cdot \exp(-\sqrt{\frac{qN_A\beta}{\varepsilon_{si}}} \cdot x)$$

$$\xi(x) = -\frac{d\Phi}{dx} = \Phi_s \cdot \sqrt{\frac{qN_A\beta}{\varepsilon_{si}}} \cdot \exp(-\sqrt{\frac{qN_A\beta}{\varepsilon_{si}}} \cdot x)$$

$$Q_s(x=0) = -\varepsilon_{si} \cdot \xi_{si} = -\sqrt{qN_A\beta\varepsilon_{si}} \cdot \Phi_s$$

$$C_{si} = \frac{\partial Q_s}{\partial \Phi_s} = -\sqrt{qN_A\beta\varepsilon_{si}}$$

$$\Rightarrow C_{si,FB} = \frac{\varepsilon_{si}}{L_D}$$

Where $$L_D = \sqrt{\frac{\varepsilon_{si}}{qN_A\beta}} = \sqrt{\frac{2\varepsilon_{si}(\frac{kT}{2q})}{qN_A}}$$

(L_D: Debye Length)

At threshold

$$\Phi_s = 2\Phi_B, p_{p0} >> n_{p0}, p_{p0} \approx N_A$$

At L.F. $n_{p0} e^{\beta \Phi_s} \approx N_A$

$$C_{si} \approx \sqrt{\frac{q\beta\varepsilon_{si}}{2}} \cdot \frac{2N_A}{\sqrt{\beta(2\Phi_B)N_A}}$$

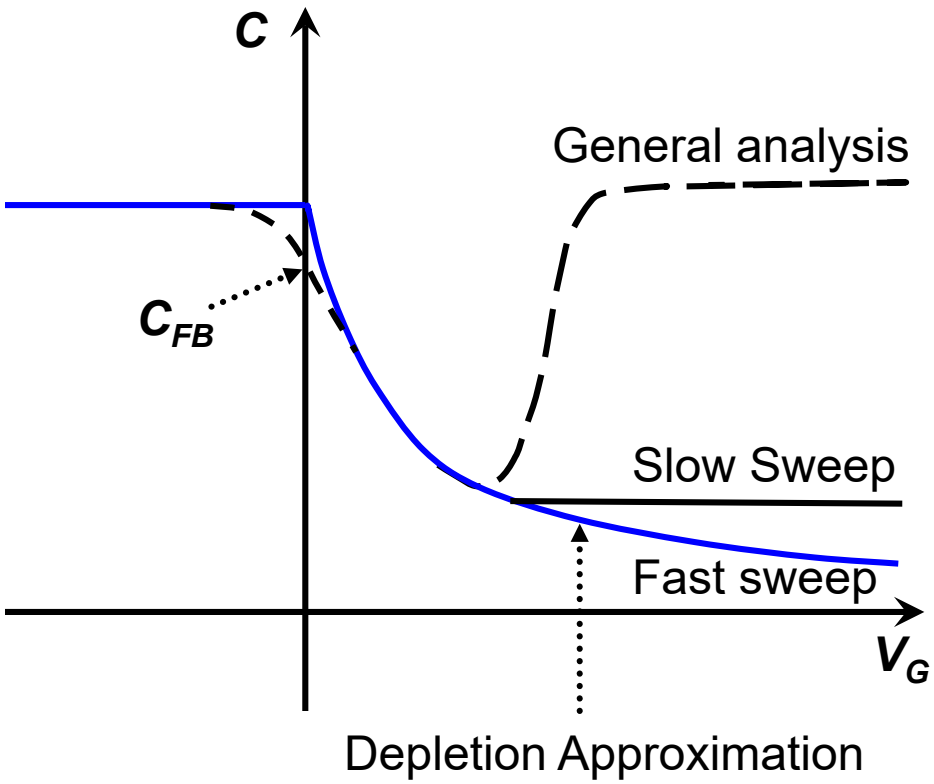
$$\Rightarrow \quad C_{si,TH} = \frac{2\varepsilon_{si}}{W_m} \quad \textbf{(L.F.)} \quad \textbf{Where} \quad W_m = \sqrt{\frac{2\varepsilon_{si} \cdot 2\Phi_B}{qN_A}}$$

However at H.F., since $n_{p0}(\beta e^{\beta \Phi_s} - \beta) \to 0$

$$C_{si} \approx \sqrt{\frac{q\beta\varepsilon_{si}}{2}} \frac{N_A}{\sqrt{\beta(2\Phi_B)N_A}}$$

$$C_{si,TH} = \frac{\varepsilon_{si}}{W_m} \quad \textbf{(H.F.)}$$

Comparison Between General Solution and Depletion Approximation



General Solution

- Consider both majority and minority carriers → general solutions of Poisson's equation yields the low frequency *C-V*

Depletion Approximation

- Good for high frequency *C-V* and deep depletion
- Not accurate for C_{FB} ← not a good approximation when *W* is small

Minimum Capacitance

$C'_{min} > C_{min}$, $V'_{min} < V_{min}(=V_T)$

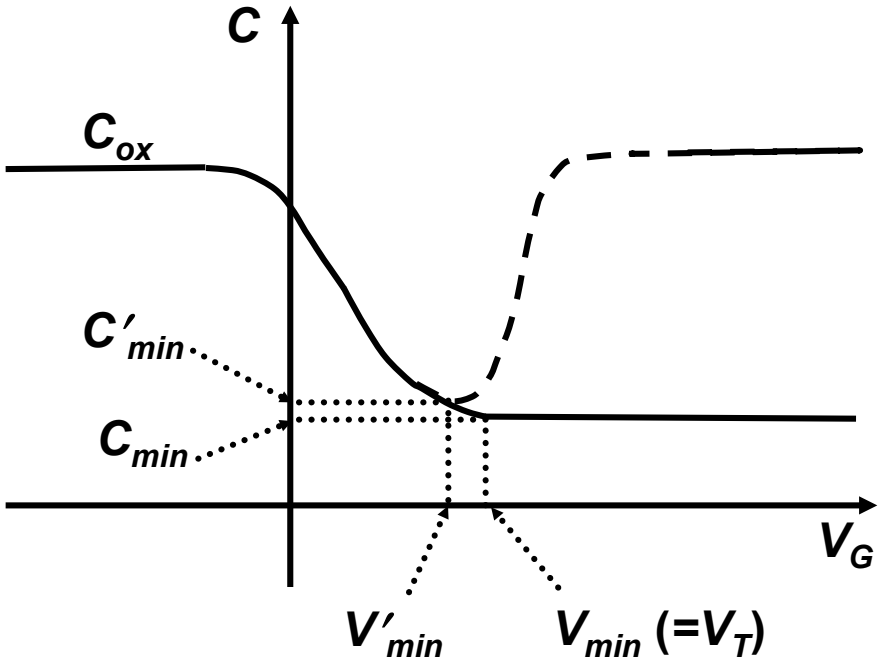
- C'_{min} can be obtained from

$$\frac{\partial C_{si}}{\partial \Phi_s} = 0$$

$$C_{si} \equiv \frac{\partial Q_{si}}{\partial \Phi_s} = \sqrt{\frac{2q\varepsilon_{si}}{\beta}} \cdot \frac{[p_{p0}(-\beta e^{-\beta\Phi_s} + \beta) + n_{p0}(\beta e^{\beta\Phi_s} - \beta)]}{2 \cdot \left[p_{p0}(e^{-\beta\Phi_s} + \beta\Phi_s - 1) + n_{p0}(e^{\beta\Phi_s} - \beta\Phi_s - 1)\right]^{\frac{1}{2}}}$$

- C_{min} can be obtained at $W=W_m$

$$\frac{C_{\min}}{C_{ox}} = \frac{C_{si(Inv.)}}{C_{ox} + C_{si(Inv.)}} = \frac{1}{1 + \frac{C_{ox}}{C_{si(Inv.)}}} = \frac{1}{1 + C_{ox} \cdot \frac{W_m}{\varepsilon_{si}}}$$



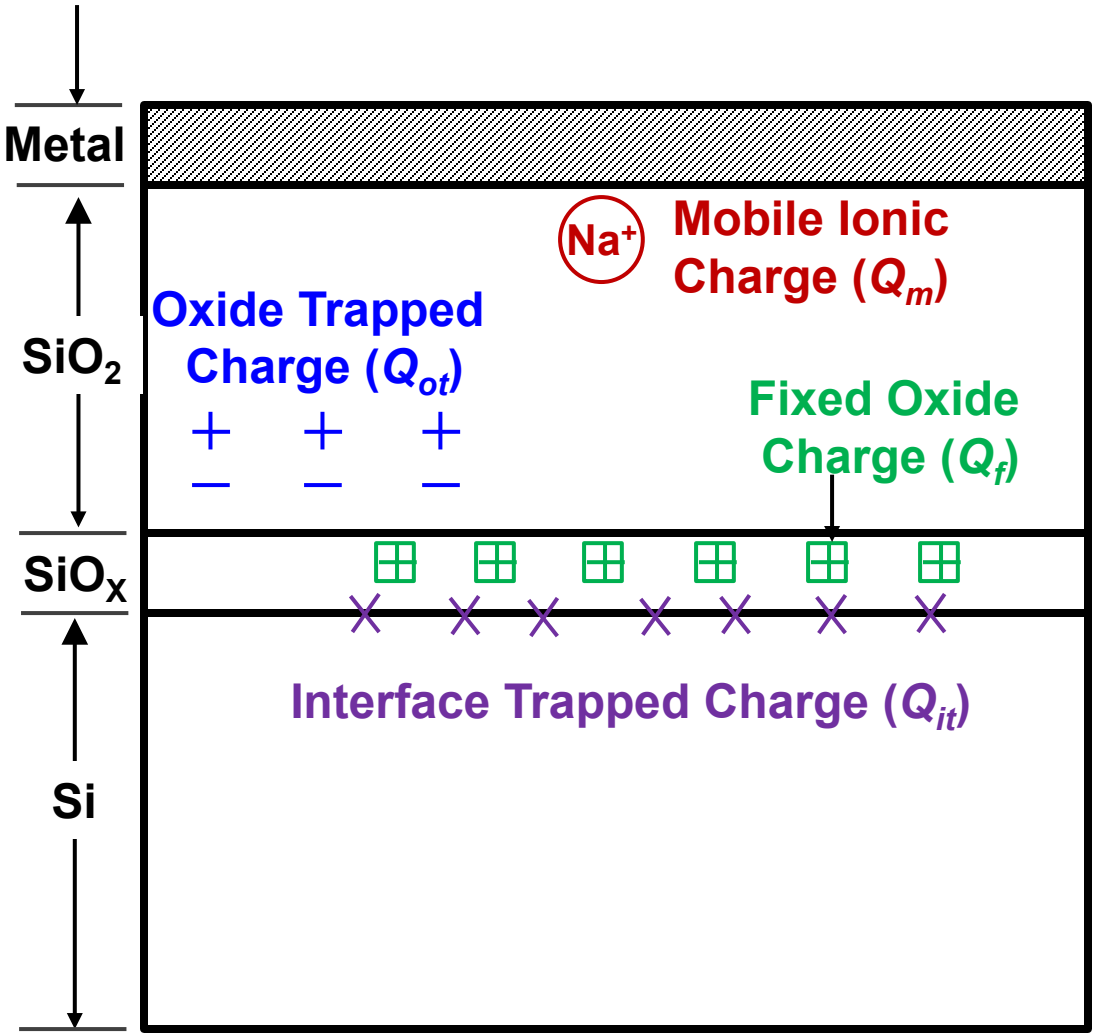
$N_A \uparrow$, then $W_m \downarrow$, thus $\frac{C_{\min}}{C_{ox}} \uparrow$

$T_{ox} \downarrow$, then $C_{ox} \uparrow$, thus $\frac{C_{\min}}{C_{ox}} \downarrow$

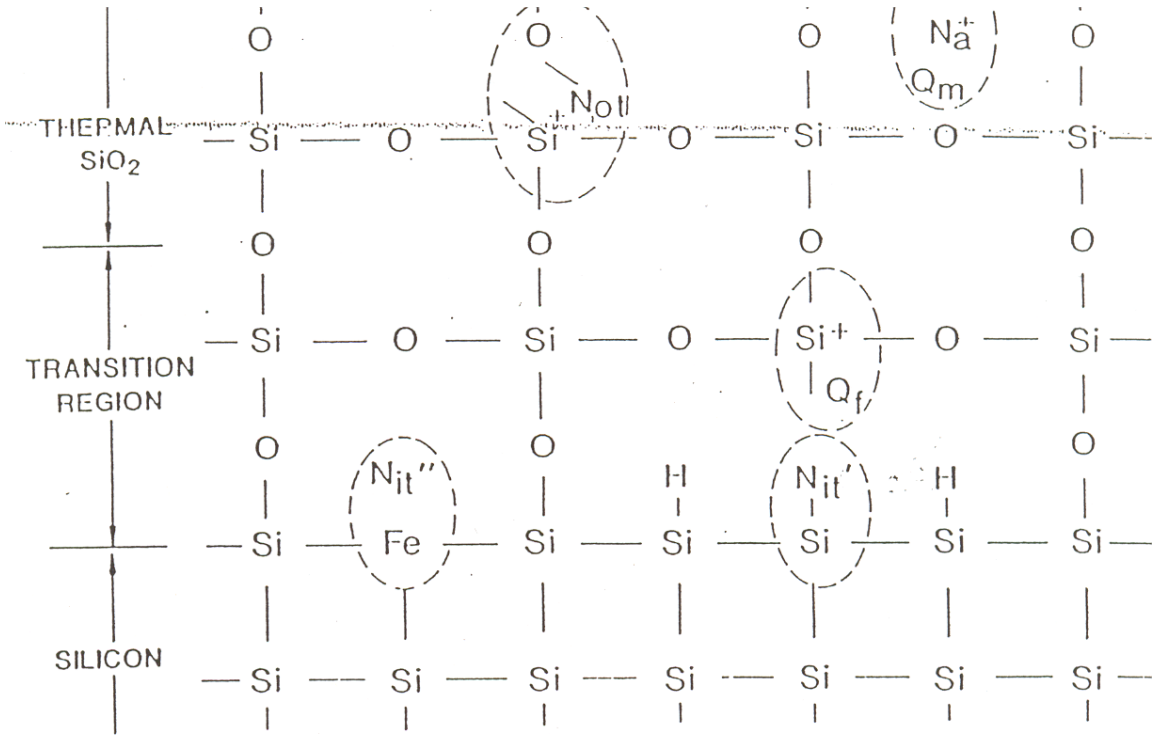
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Defect Charges in MOS



Atomic Configurations and Defects near Si/SiO_2



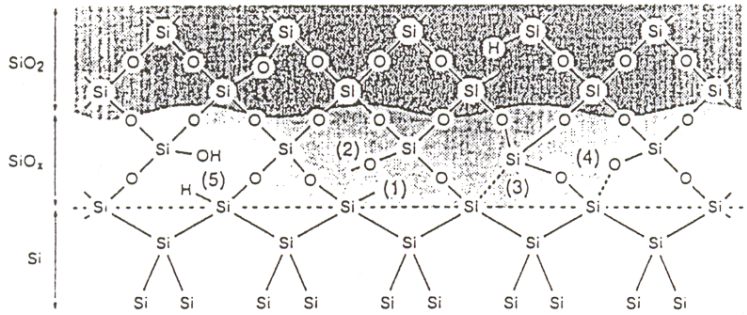


Illustration of possible atomic configurations and defects near the Si-SiO_2 inter-

(1) Silicon dangling bond: may act as hole trap

(2) Non-bridging oxygen: formed from broken Si-O bond

(3) Weak Si-Si bond: electrically active when broken

(4) Strained Si-O bond: break of perfect degree (109°) by stress

(5) Hydrogen-containing species: good for curing D_{it}, but bonds are too weak to endure electrical stress

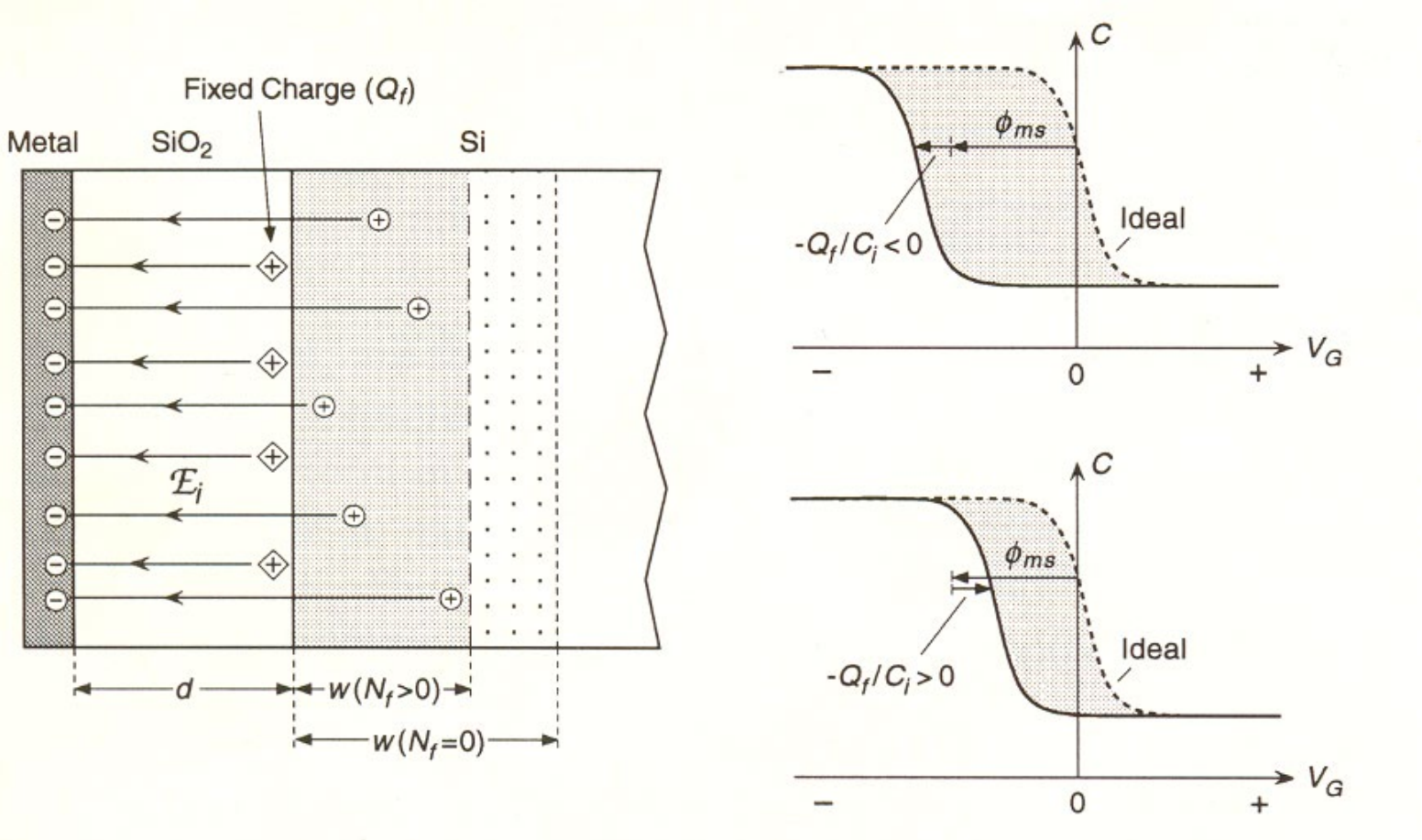
Interface Trapped Charge (Q_{it})

- ORIGINATED from structural defect (dangling bonds) related to the oxidation process, metallic impurities, or bond breaking processes such as X-ray irradiation and hot carrier injection.
- LOCATED at the Si-SiO_2 interface with energy states in the silicon forbidden bandgap and which can exchange charges with silicon in a short time.
- CAUSES degradation of transconductance and subthreshold characteristics of MOSFET.
- TESTED by *C-V* measurement.
- REDUCED by Forming Gas Anneal (H_2/N_2).
- COMMON CRITERIA: $D_{it} < 10^{11}\ eV^{-1}{\cdot}cm^{-2}$.

Fixed Oxide Charge (Q_f)

- ORIGINATED from Si dangling bonds due to the non-stoichiometric region (strained region) of SiO_2 near the Si-SiO_2 interface (SiO_x where $1 < x < 2$).
- LOCATED roughly within 10 Å of the Si interface, immobile; cannot exchange charges unlike Q_{it}.
- CAUSES V_{FB} and V_T shifts, and mobility degradations.
- TESTED by *C-V* measurement.
- REDUCED by high-temperature post-oxidation anneal.
- COMMON CRITERIA: $Q_f = 10^{10}$ cm^{-2} ~ 10^{12} $cm^{-2,}$ depending on annealing condition and Si-wafer orientation.

Effect of Q_f on MOS Capacitor



For n-type substrate

For p-type substrate

$$V_{FB} = \Phi_{MS} - \frac{Q_f}{C_{ox}}$$

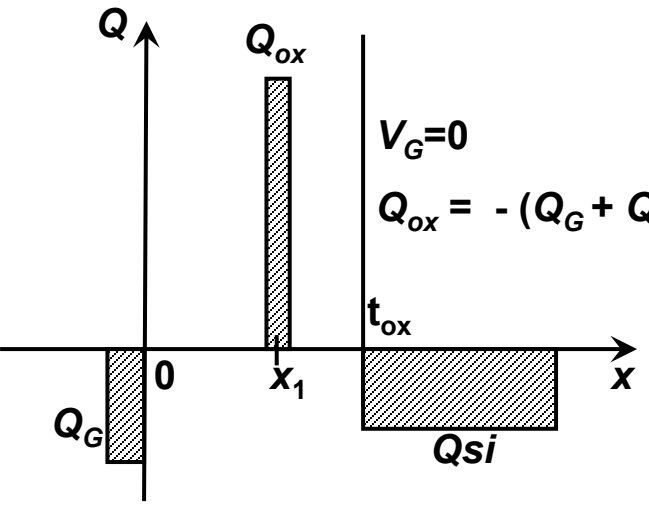
Oxide Trapped Charge (Q_{ot})

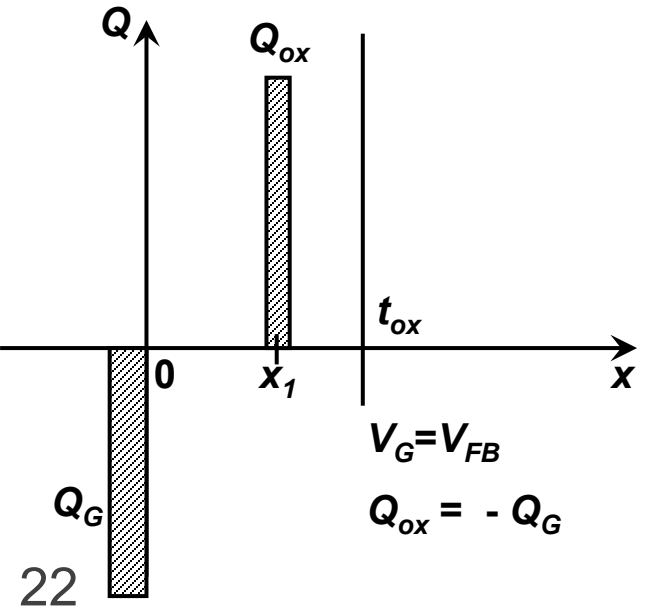
- Believed to be ORIGINATED from H- and OH- bonds.
- LOCATED throughout the bulk oxide.
- CAUSES oxide breakdown and V_T instability.
- TESTED by TDDB high-field stress.
- REDUCED by using optimized oxidation and annealing processes.
- COMMON CRITERIA: $Q_{ot} = 10^{9}$ cm^{-2} ~10^{13} cm^{-2}.

Mobile Ionic Charge (Q_M)

- ORIGINATED from alkali ions (Na^+, K^+) and heavy metal contamination
- LOCATED throughout the MOS
- CAUSES V_{FB} instability
- TESTED by bias-temperature stress
- REDUCED by careful management of contamination sources
- COMMON CRITERIA: $Q_M < 10^{10}$ cm^{-2} for present-day ULSI technology

V_{FB} Shift due to Oxide Charges





At $V_G \approx V_{FB}$

$$\xi_{ox} \approx -\frac{Q_{ox}}{\varepsilon_{ox}} \qquad (0<x<x_1)$$

$$\Delta V_{FB} = x_1 \xi_{ox} = -\frac{x_1}{\varepsilon_{ox}} Q_{ox} = -\frac{x_1}{C_{ox} \cdot t_{ox}} Q_{ox} \qquad (0<x_1<t_{ox})$$

In general, $$\Delta V_{FB} = -\frac{1}{C_{ox} \cdot t_{ox}} \int_0^{t_{ox}} x\rho(x)dx$$

Assuming Q_f' **at** $x_1 = t_{ox}$

$$\Delta V_{FB} = -\frac{Q_f'}{C_{ox}} - \frac{1}{C_{ox} \cdot t_{ox}} \int_0^{t_{ox}} x\rho(x)dx$$

$$\therefore V_{FB} = \Phi_{MS} - \frac{Q_f'}{C_{ox}} - \frac{1}{C_{ox} \cdot t_{ox}} \int_0^{t_{ox}} x\rho(x)dx$$

$$V_T = V_{FB} + 2\Phi_B + \frac{|Q_{si}|}{C_{ox}}$$

$$= \Phi_{MS} - \frac{Q_f'}{C_{ox}} - \frac{1}{C_{ox} \cdot t_{ox}} \int_0^{t_{ox}} x\rho(x)dx + \frac{2kT}{q} \ln \frac{N_A}{n_i} + \frac{1}{C_{ox}} \sqrt{4qN_A\varepsilon_{si}\Phi_B}$$

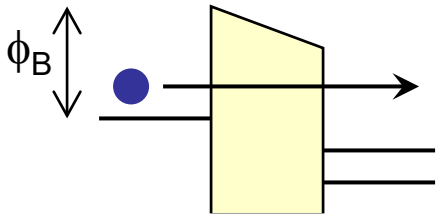
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Carrier Transport in Gate Dielectric Films

- **Direct Tunneling**: Quantum mechanical tunneling through gate oxide

$$J = \frac{q^2}{2\pi h d^2}\left[\left(\Phi_B - \frac{V}{2}\right)\cdot\exp\left(-\frac{4\pi d\sqrt{2m^* q}}{h}\sqrt{\Phi_B - \frac{V}{2}}\right) - \left(\Phi_B + \frac{V}{2}\right)\cdot\exp\left(-\frac{4\pi d\sqrt{2m^* q}}{h}\sqrt{\Phi_B + \frac{V}{2}}\right)\right]$$

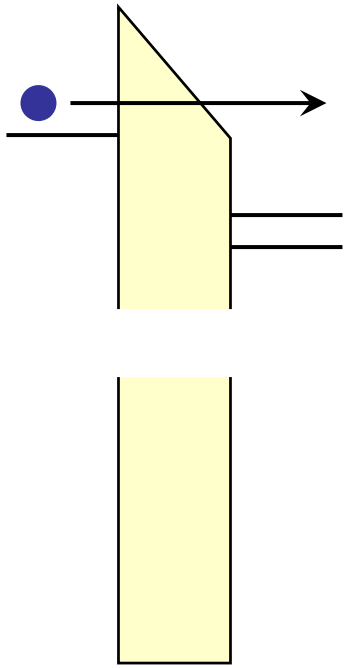


m:* effective mass of electron

h: Plank's constant

- **Fowler-Nordheim Tunneling**: Field assisted tunneling at high electric field

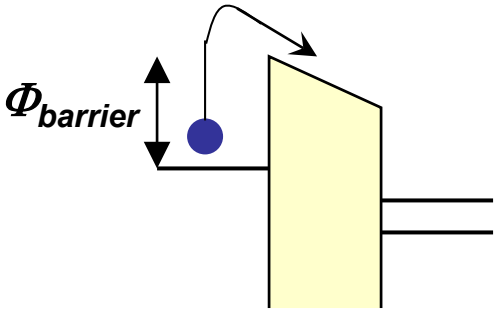
$$J = \frac{q^2}{8\pi h \Phi_B}\xi_i^2 \cdot \exp\left(-\frac{2\Phi_B^{\frac{2}{3}}}{3\xi_i}\right)$$



Carrier Transport in Gate Dielectric Films

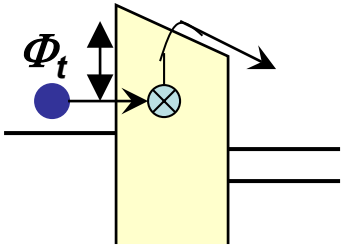
- **Schottky Emission**: Thermionic Emission across the Barrier Height

$$J = A^{*}T^{2}\exp\left[\frac{-q\left(\Phi_B - \sqrt{q\xi_i / 4\pi\varepsilon_i}\right)}{kT}\right],$$ A^{*}: Richardson Constant



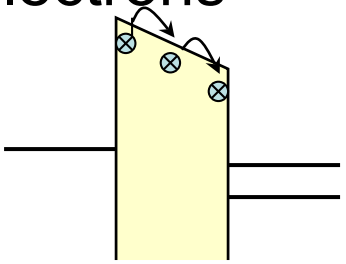
- **Poole-Frenkel Emission**: Field-enhanced Thermal Excitation of Trapped Electrons

$$J \propto \xi_i \cdot \exp\left[\frac{-q\left(\Phi_t - \sqrt{q\xi_i / \pi\varepsilon_i}\right)}{kT}\right]$$

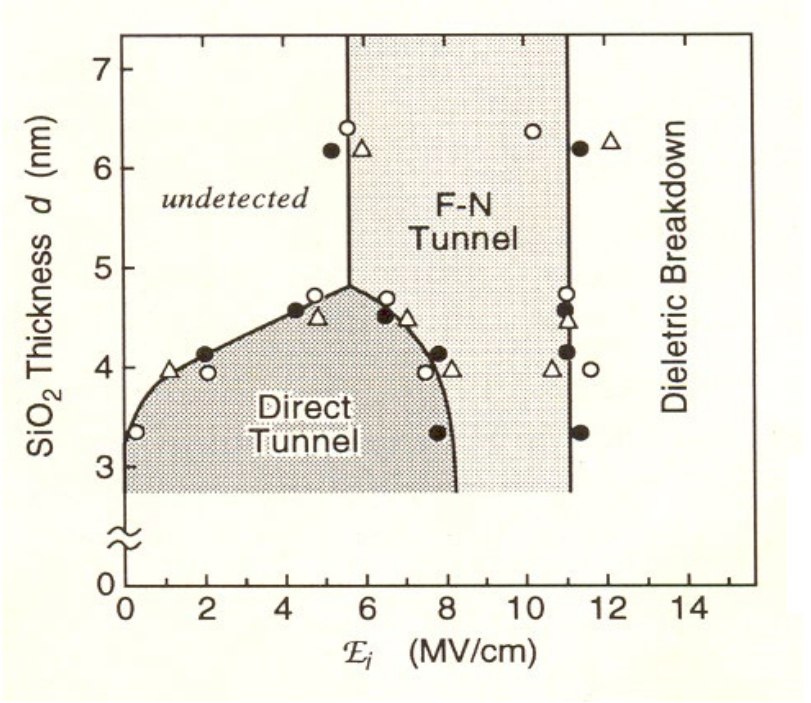


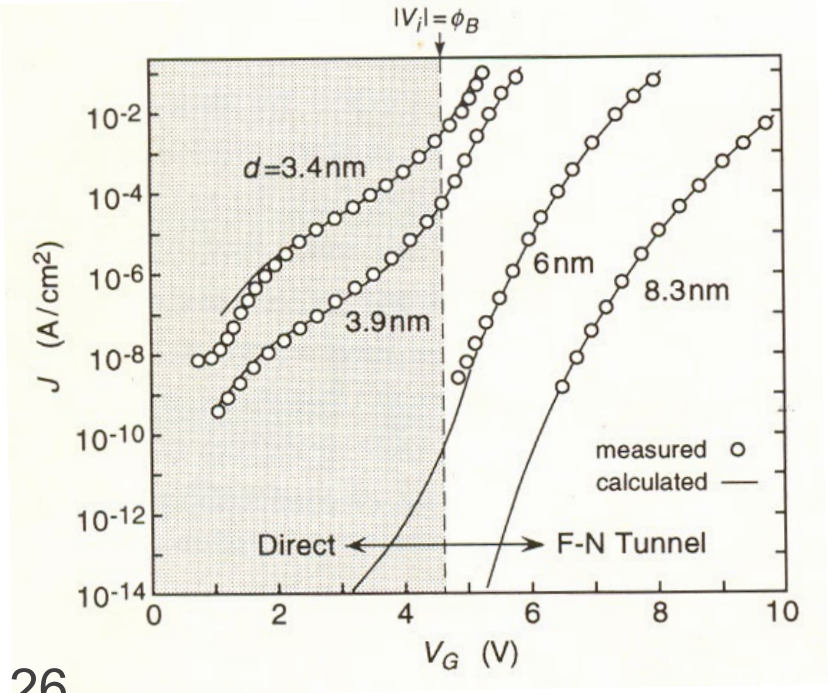
- **Ohmic Conduction**: Trap-assisted Hopping of Electrons

$$J \propto \xi_i \cdot \exp\left[\frac{-\Delta E_{ae}}{kT}\right],$$ E_{ae}: Activation Energy of Electron



Direct & F-N tunneling: Conduction Processes in ultrathin SiO_2 as a function of ξ_i (applied gate bias) and T_{ox}





Fowler-Nordheim Plot

